

BAYINA ROHIT

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Education

Vellore Institute of Technology

Integrated MTech in Artificial Intelligence | CGPA: 9.02/10

Sep 2022 – Present

Expected Graduation: 2027

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL

AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, RAG, LangChain, LangGraph, FastAPI, Vector Databases

Web & Backend: Node.js, Express, React, MongoDB

Tools & Platforms: AWS, Git, GitHub, Docker, JupyterLab, Postman

Experience

Machine Learning Intern

Gravitas corp

Nov 2025

- Built a multimodal AI application supporting voice, text, and image inputs using LLaMA and LlamaIndex, optimizing inference workflows to reduce end-to-end latency by 40%.
- Developed a multi-agent workflow using Python and LlamaIndex to coordinate automated tasks, improving execution reliability and increasing user session retention by 35%.
- Implemented web automation scripts using Playwright and OAuth2 authentication to interact with 10+ protected services, achieving 98% task success through improved error handling.
- Collaborated with engineers to debug runtime failures, analyze system performance, and improve reliability through structured testing and logging.

Projects

Panoptic Image Segmentation— Detectron2, Deep Learning, OpenCV

May 2023

- Implemented a panoptic image segmentation model using Detectron2 and OpenCV to perform unified semantic and instance segmentation, achieving 0.91 mean IoU on validation datasets
- Collected and annotated a dataset of 500+ images to train and evaluate segmentation models, ensuring consistent labeling and balanced object distribution.
- Evaluated model performance using multiple metrics including accuracy and IoU to analyze segmentation quality and identify failure cases.

BenchForge — MERN Stack , PyTorch, TensorFlow, Docker, OpenCV, Scikit-learn

Feb 2024

- Designed an AI benchmarking tool using a MERN stack interface and Python evaluation scripts to measure performance across 10+ deep learning models.
- Measured model performance including inference latency, accuracy, CPU/GPU utilization, and memory usage across multiple hardware configurations.
- Improved system throughput by enabling concurrent evaluation jobs, increasing benchmarking efficiency by 35%.

AURA - Service Center Dashboard — AI APIs, Recharts, React, TypeScript, CSS, Vite

Oct 2025

- Engineered a vehicle health monitoring dashboard using React and TypeScript to visualize operational KPIs and real-time alerts across multiple modules.
- Integrated outputs from multiple machine learning pipelines into the dashboard to display real-time confidence scores and system alerts.
- Executed efficient state management and data flow to support frequent real-time updates while maintaining stable UI performance.

Certifications

- **Applied Machine Learning in Python** – University of Michigan (Coursera)
- **Web Development with MERN Stack by FACE Prep** focused on full-stack application development.
- **Apna College** – **DSA with Java** covering arrays, linked lists, stacks, queues, trees, graphs, recursion, and DP.